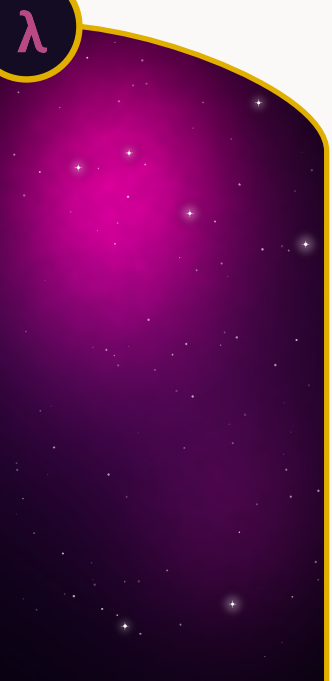


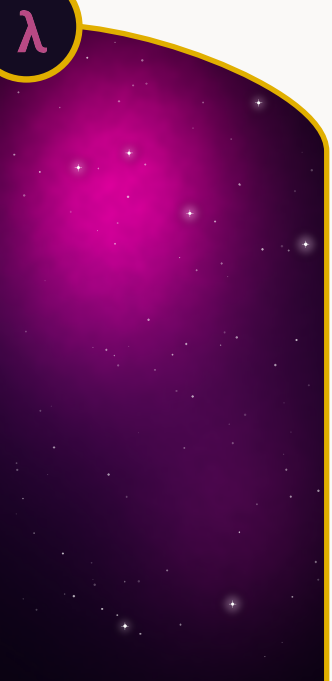
L3A: Higher-order Functions; Scope of Names

CS1101S: Programming Methodology

Boyd Anderson

26th August, 2026

Version 1.1 — Built: 2026-08-26 05:25:29



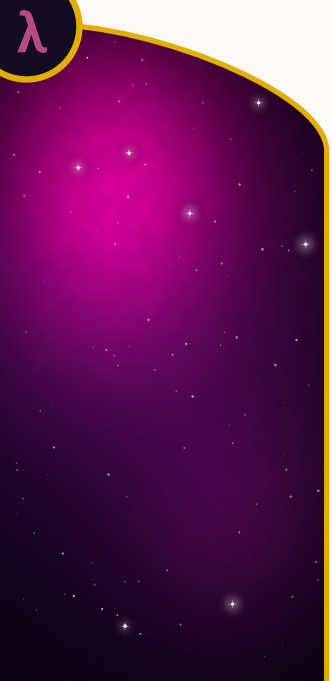
Announcements

Nested declaration assignments

More fun with recursion

Higher-order functions

Scope of names



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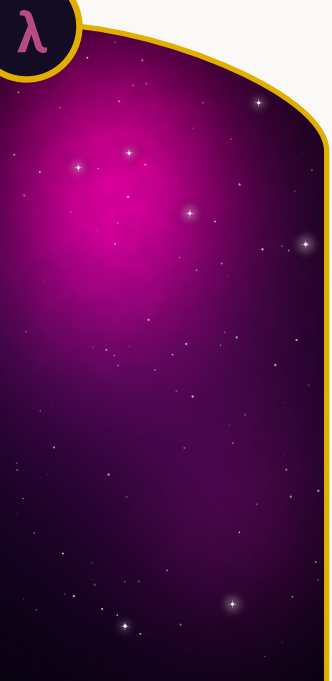

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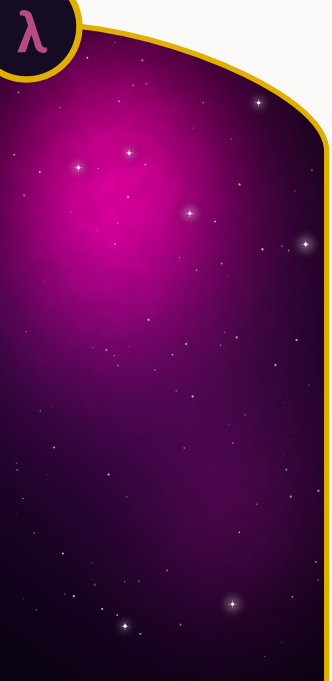
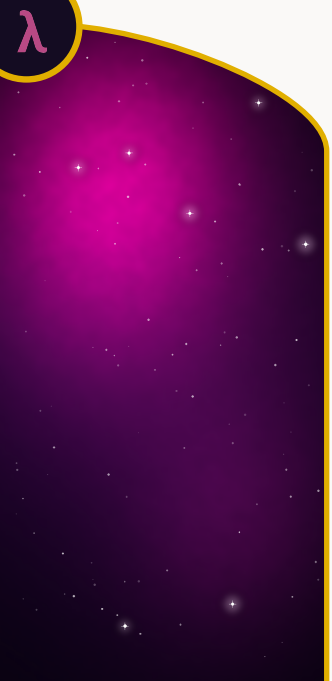
- **Forum Hours**. Every Tuesday 6:30pm-8:30pm. Venue is COM1 basement.
- Reading Assessment 1: Read our announcement carefully. Sample questions have been uploaded.



Some Words of Advice



Some Words of Advice

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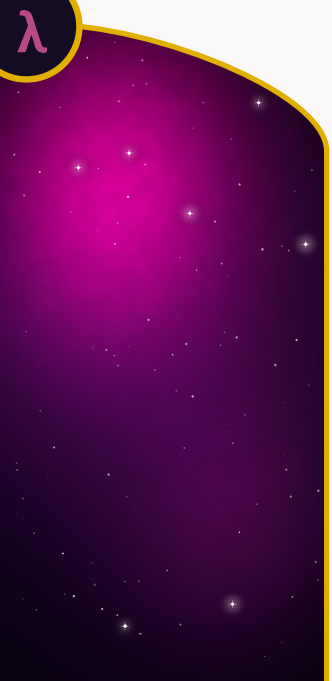
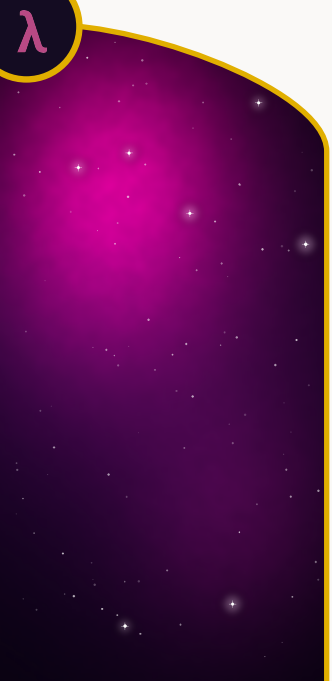


Some Words of Advice

- Read the textbook
- Use the substitution model (and stepper tool, if needed)



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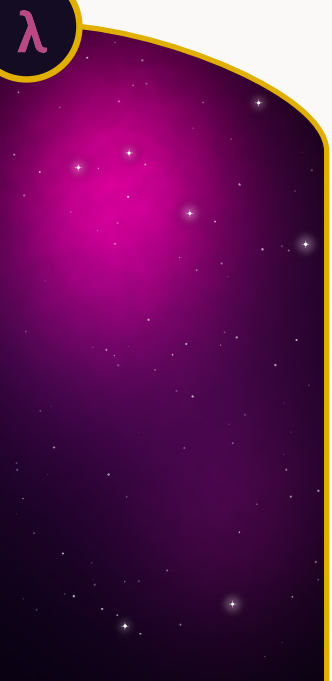
- Think, then program



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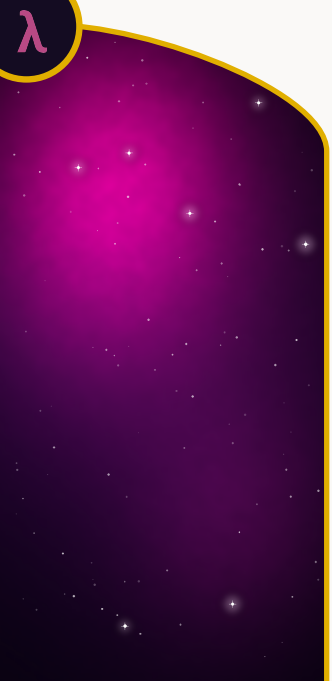
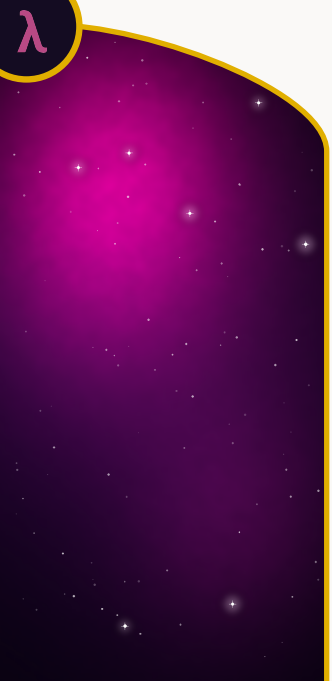
- Read the textbook
- Use the substitution model (and stepper tool, if needed)
- Think, then program
- Less is more
- Use AI smartly (to optimise your learning)





Names can refer to intermediate values

Example from SICPy 1.3.2

$$f(x, y) = x(1 + xy)^2 + y(1 - y) + (1 + xy)(1 - y)$$

Compute $f(2, 3)$

Names can refer to intermediate values

Example from SICPy 1.3.2

$$f(x, y) = x(1 + xy)^2 + y(1 - y) + (1 + xy)(1 - y)$$

Compute $f(2, 3)$

```
def f(x, y):
    return (x * square(1 + x * y)
            + y * (1 - y)
            + (1 + x * y) * (1 - y))

print(f(2, 3))
```

Names can refer to intermediate values

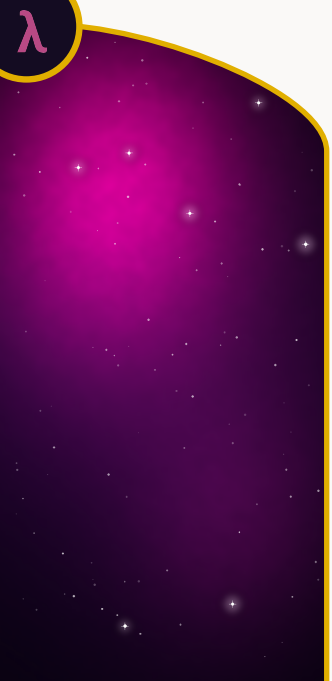
Example from SICPy 1.3.2

$$f(x, y) = x(1 + xy)^2 + y(1 - y) + (1 + xy)(1 - y)$$

Compute $f(2, 3)$

```
def f(x, y):
    a = 1 + x * y
    b = 1 - y
    return x * square(a) + y * b + a * b

print(f(2, 3))
```



Announcements

Nested declaration assignments

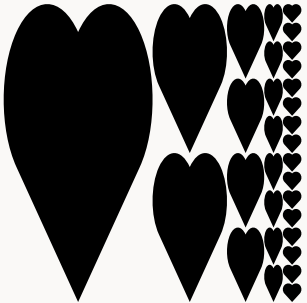
More fun with recursion
- Example: fractal runes ("Rune Reading")
- Conditional statements (1.3.2)
- Example: coin change (1.2.2)

Higher-order functions

Scope of names

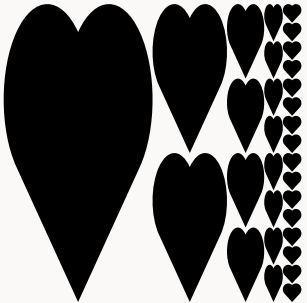
Example from Mission “Rune Reading”: `fractal`

Define a function `fractal` that returns pictures like this:



Example from Mission “Rune Reading”: `fractal`

Define a function `fractal` that returns pictures like this:

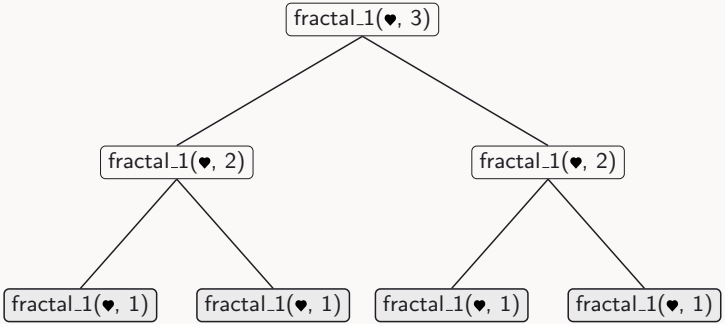


when we call it like this: `fractal(heart, 5)`

“Rune Reading” Example: `fractal`, Solution 1

```
def fractal_1(rune, n):
    return (rune if n == 1
            else beside(rune,
                stack(fractal_1(rune, n - 1),
                      fractal_1(rune, n - 1))))
```

The Call Tree for `fractal_1(heart, 3)`



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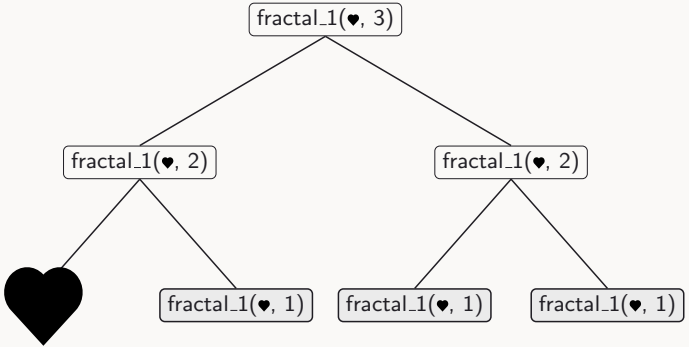

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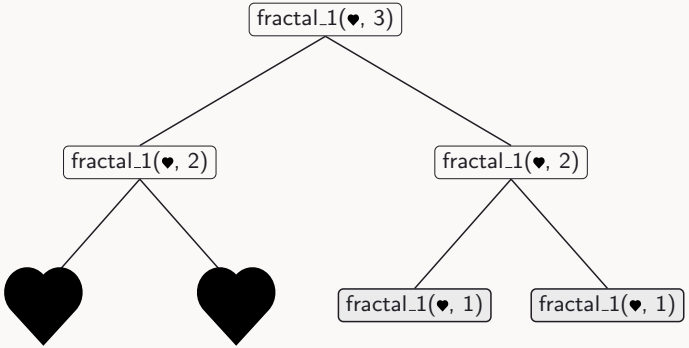

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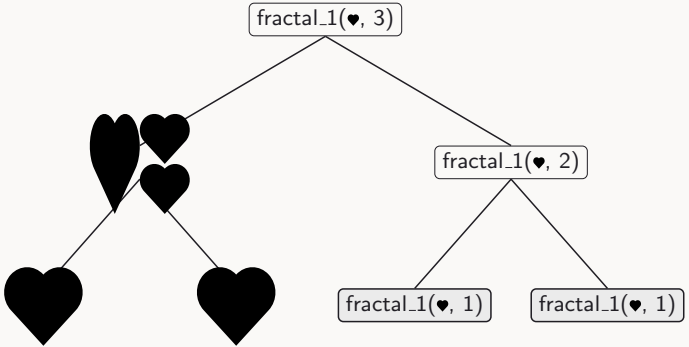

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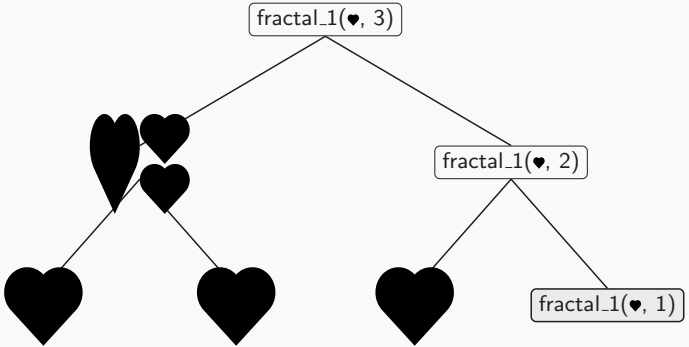

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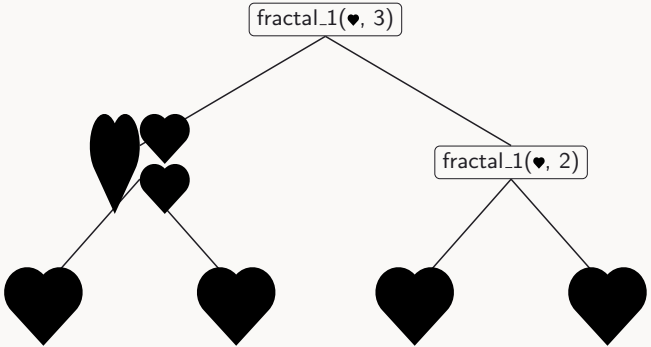

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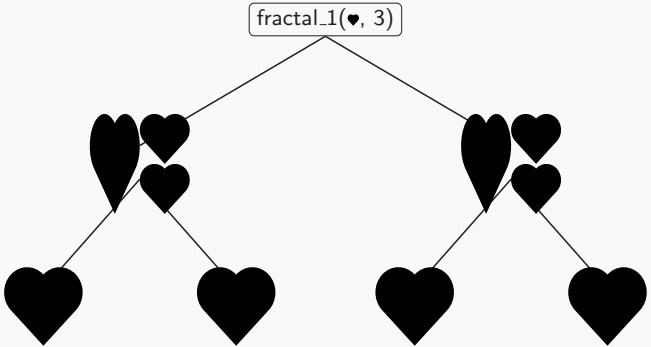

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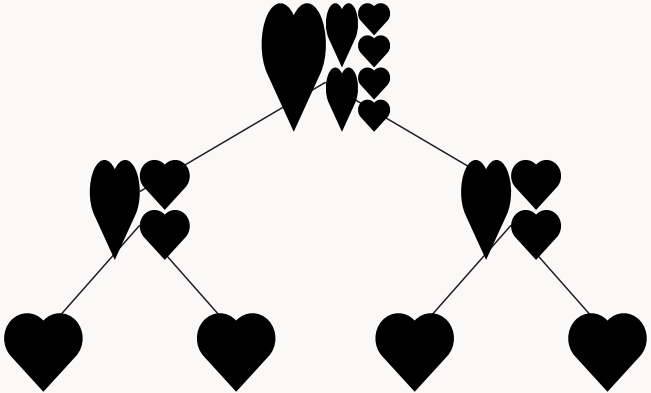
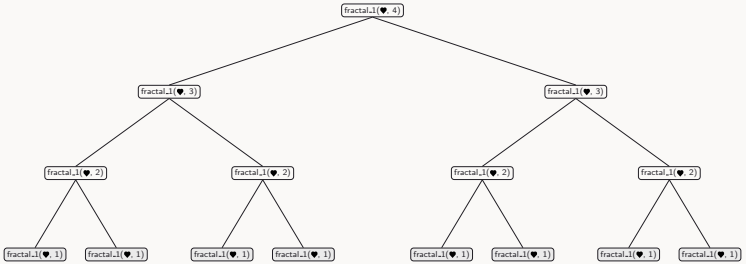


The Call Tree for `fractal_1(heart, 4)`



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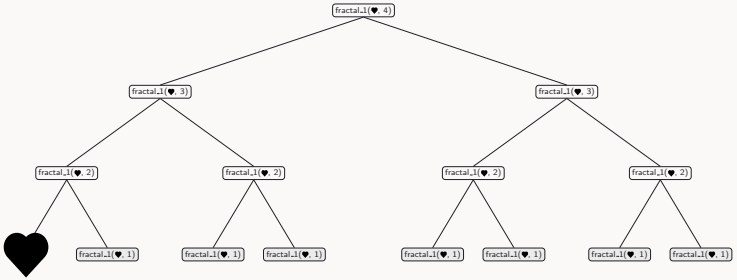

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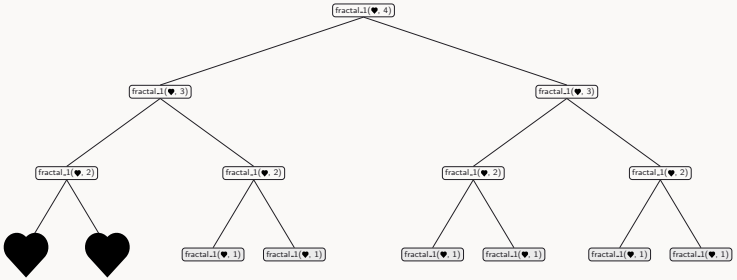

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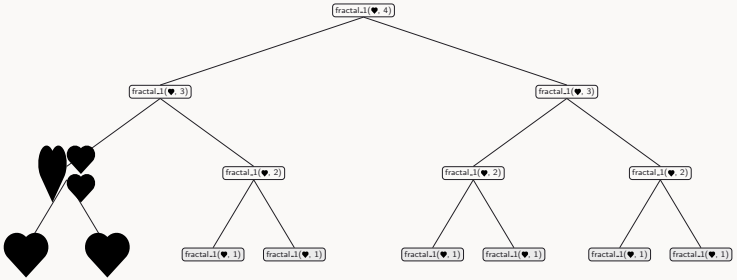

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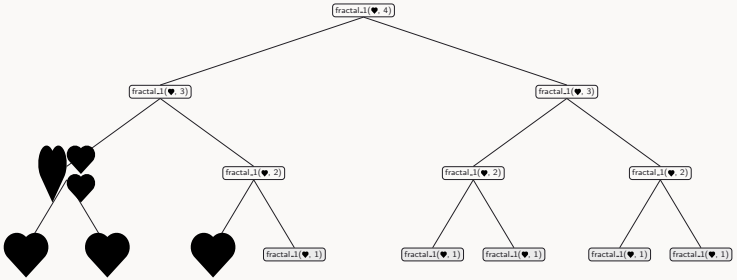

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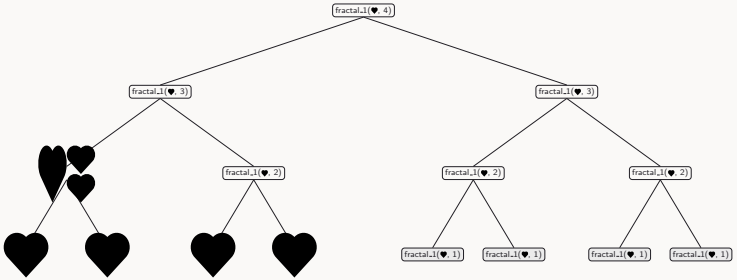

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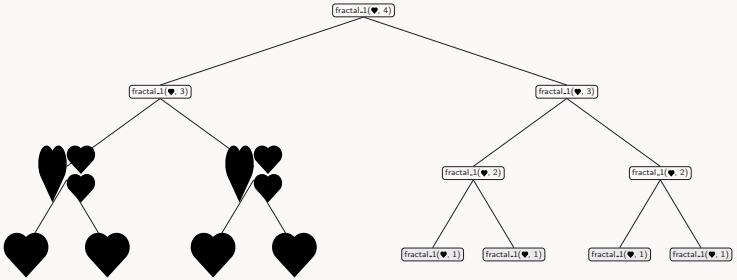

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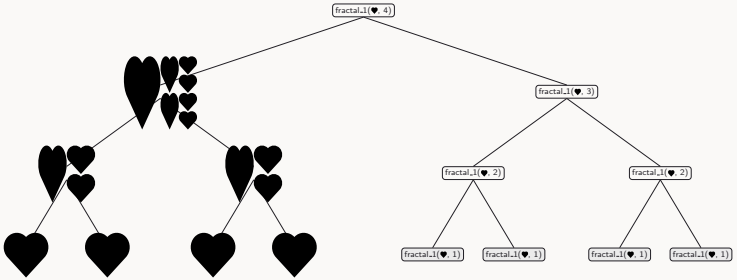

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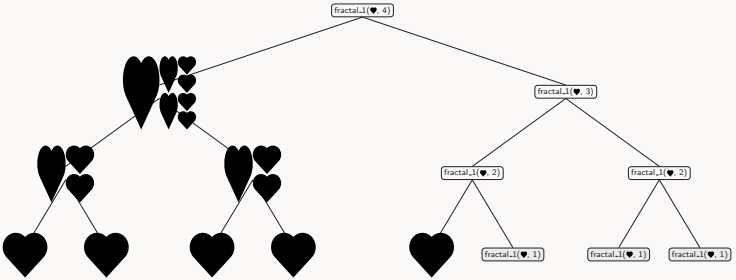

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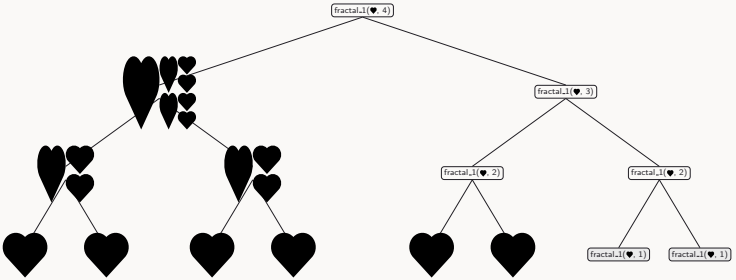

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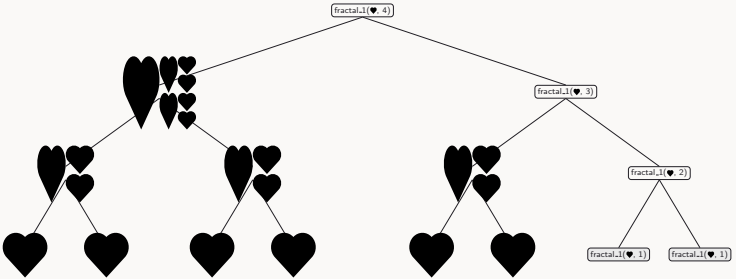

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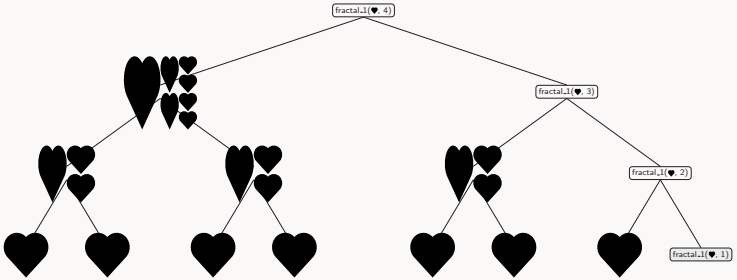

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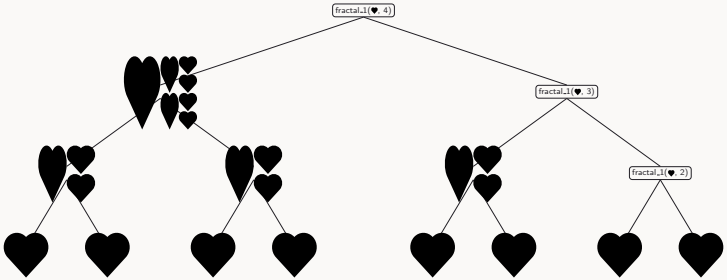

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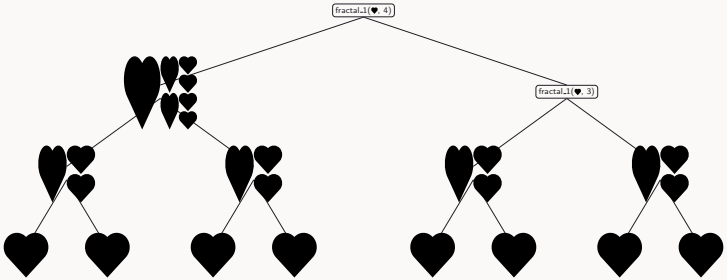

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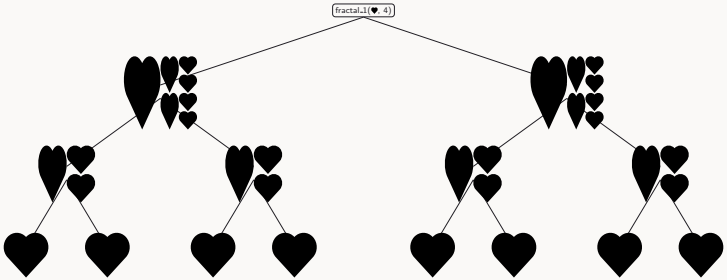

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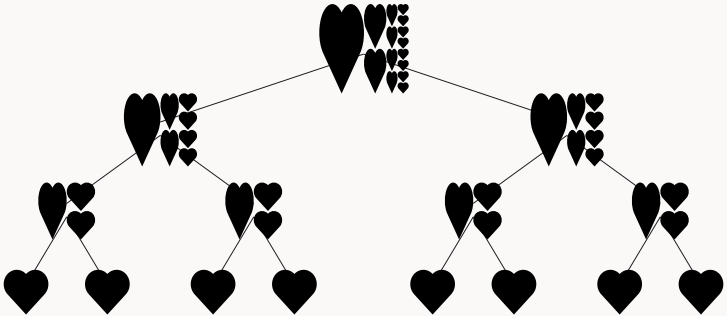

Can we avoid tree recursion?

```
def fractal_1(rune, n):
    return (rune if n == 1
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```

Question

Can we implement this function with linear recursion?

Is this a good idea?

```
def fractal_2(rune, n):
    smaller_frac = fractal_2(rune, n - 1)
    return (rune if n == 1
            else beside(rune,
                stack(smaller_frac, smaller_frac)))
```

Is this a good idea?

```
def fractal_2(rune, n):
    smaller_frac = fractal_2(rune, n - 1)
    return (rune if n == 1
            else beside(rune,
                stack(smaller_frac, smaller_frac)))
```

Can we declare a name...

Is this a good idea?

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def fractal_2(rune, n):
    smaller_frac = fractal_2(rune, n - 1)
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                stack(smaller_frac, smaller_frac)))
```

Can we declare a name...

...**just for the alternative of the conditional**?

Conditional statements (see SICPy 1.3.2)

```
def fractal_3(rune, n):
    if n == 1:
        return rune
    else:
        f = fractal_3(rune, n - 1)
        return beside(rune, stack(f, f))
```

Conditional statements (see SICPy 1.3.2)

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- Each branch of the conditional can have local declarations.

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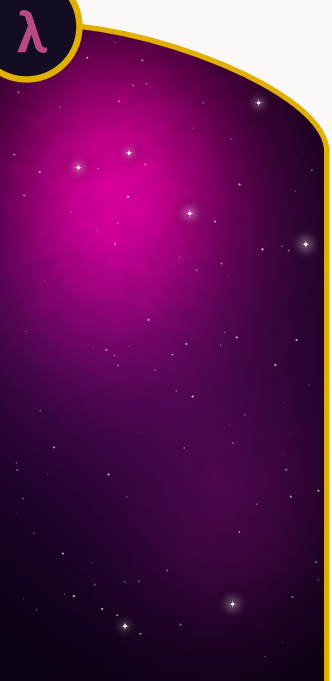
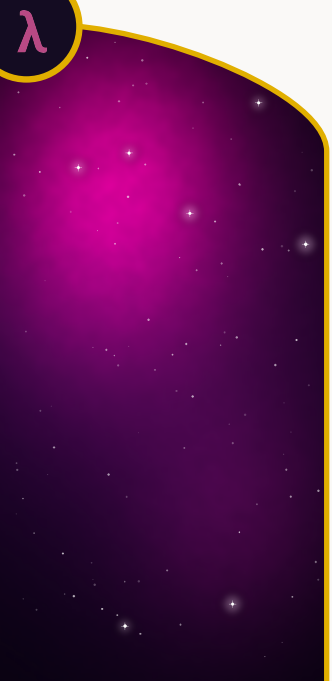
- Each branch of the conditional can have local declarations.
- A declaration assignment in a branch is only evaluated when the branch actually runs (more on scope shortly).
- Remember to return a result *in each branch*.
  (Otherwise `None` is returned.)

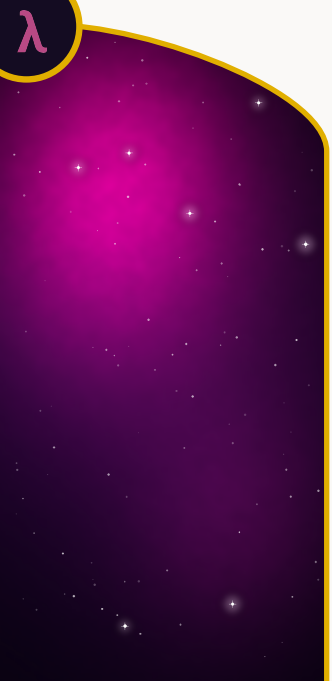


Example: coin change (1.2.2)



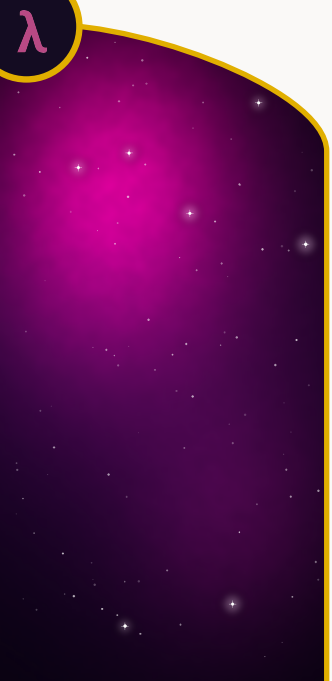
Example: coin change (1.2.2)

- Given: Different kinds of coins (unlimited supply)



Example: coin change (1.2.2)

- Given: Different kinds of coins (unlimited supply)
- Given: Amount of money in cents



Example: coin change (1.2.2)

- Given: Different kinds of coins (unlimited supply)
- Given: Amount of money in cents
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Step 1

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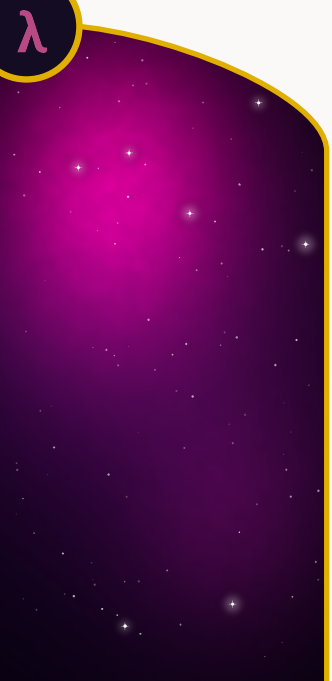
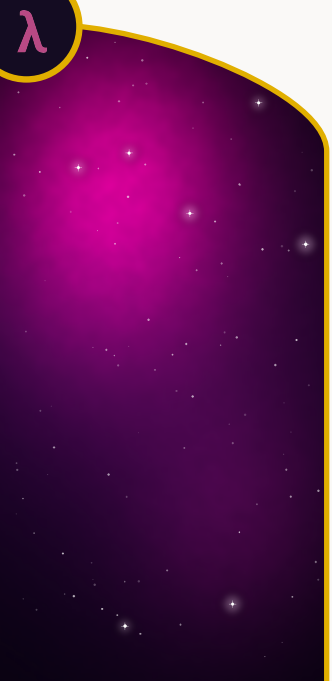


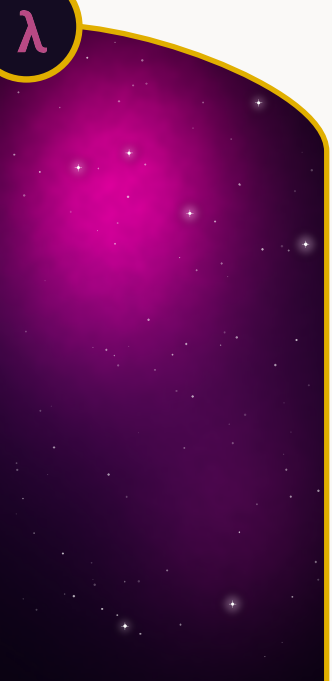
Play

- Given: Different kinds of coins (unlimited supply)
- Given: Amount of money in cents
- Wanted: Number of ways to change amount into coins



Play

- Given: Different kinds of coins (unlimited supply)
- Given: Amount of money in cents
- Wanted: Number of ways to change amount into coins
- How about 10 cents?



Play

- Given: Different kinds of coins (unlimited supply)
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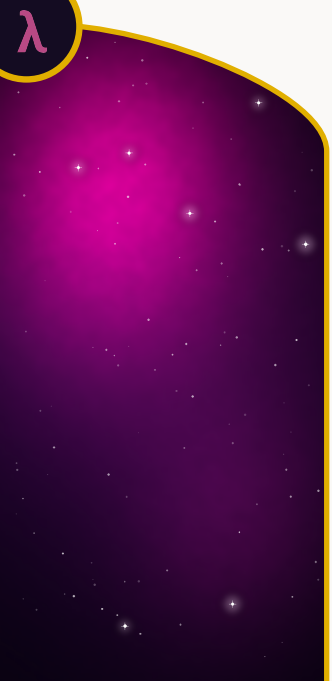
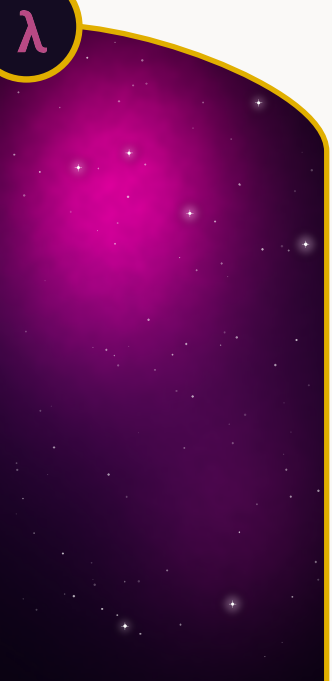
- Wanted: Number of ways to change amount into coins
- How about 10 cents?
- How about 20 cents?



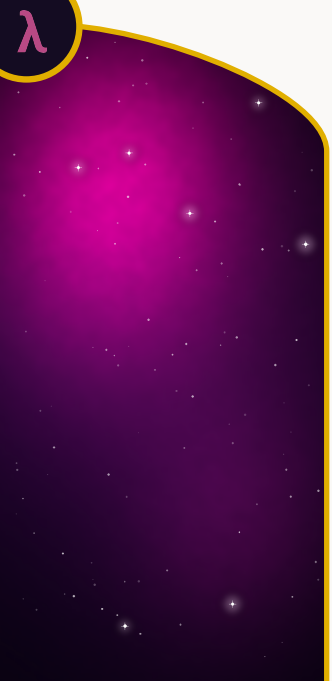
Play

- Given: Different kinds of coins (unlimited supply)
- Given: Amount of money in cents
- Wanted: Number of ways to change amount into coins
- How about 10 cents?
- How about 20 cents?
- 5 cents?



Play

- Given: Different kinds of coins (unlimited supply)
- Given: Amount of money in cents
- Wanted: Number of ways to change amount into coins
- How about 10 cents?
- How about 20 cents?
- 5 cents?
- 0 cents?



Play

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- 5 cents?
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Play

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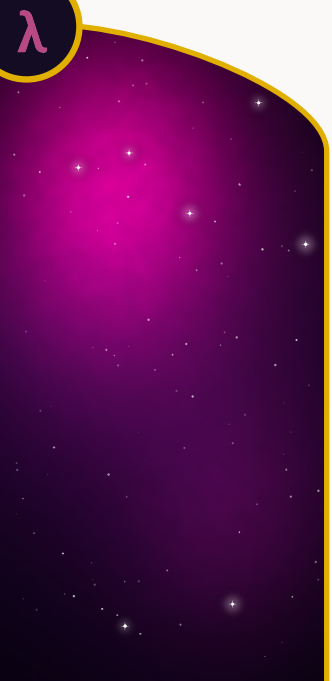
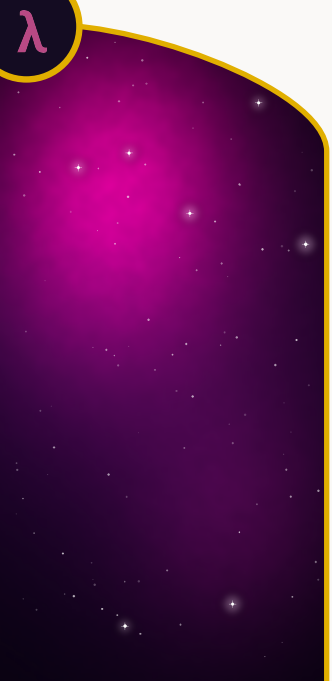
Step 2

Play with examples

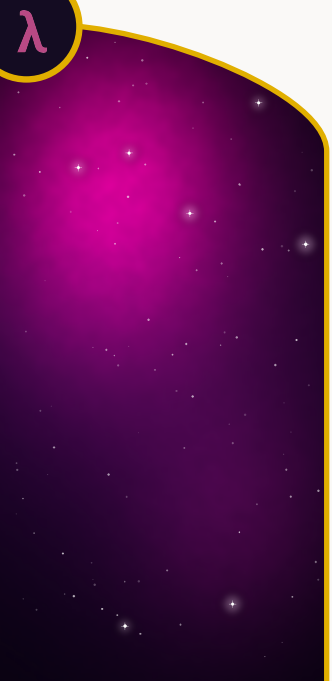


Think



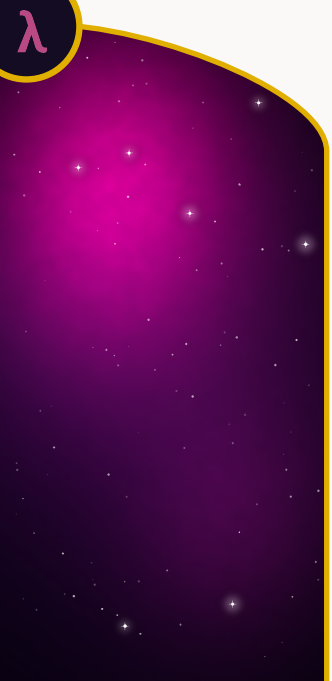
Think

- Example: Number of ways to change 120 cents



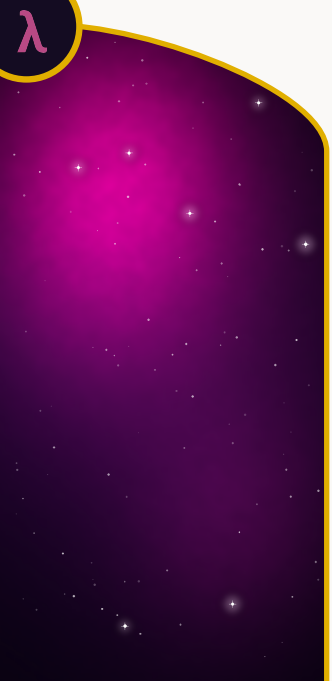
Think

- Example: Number of ways to change 120 cents
- Idea: Highest coin is either 100 or not 100



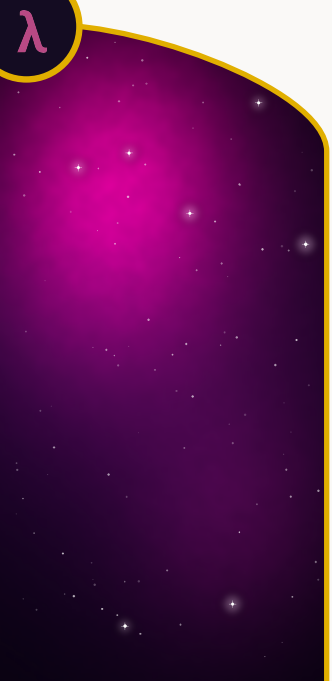
Think

- Example: Number of ways to change 120 cents
- Idea: Highest coin is either 100 or not 100
- In the first case: Smaller problem



Think

- Example: Number of ways to change 120 cents
- Idea: Highest coin is either 100 or not 100
- In the first case: Smaller problem
- In the second case: Smaller problem



Think

- Example: Number of ways to change 120 cents
- Idea: Highest coin is either 100 or not 100
- In the first case: Smaller problem
- In the second case: Smaller problem
- Base cases?

Think

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- In the first case: Smaller problem
- In the second case: Smaller problem
- Base cases?

Step 3

Think: existing solution? divide-and-conquer? “wishful thinking”?

Representing the kinds of coins

```
def first_denomination(kinds_of_coins):
    return (5 if kinds_of_coins == 1
            else 10 if kinds_of_coins == 2
            else 20 if kinds_of_coins == 3
            else 50 if kinds_of_coins == 4
            else 100 if kinds_of_coins == 5
            else 0)
```

Idea in Python

```
def cc(amount, kinds_of_coins):
    return (...                        # base cases
            if ...
            else cc(amount -
    first_denomination(kinds_of_coins),
                       kinds_of_coins)
                  + cc(amount, kinds_of_coins - 1))
```

Idea in Python

```
def cc(amount, kinds_of_coins):
    return (...                          # base cases
            if ...
            else cc(amount -
    first_denomination(kinds_of_coins),
                        kinds_of_coins)
                   + cc(amount, kinds_of_coins - 1))
```

Step 4

Program using Python

Adding base cases

```
def cc(amount, kinds_of_coins):
    return (1 if amount == 0
            else 0 if amount < 0 or kinds_of_coins == 0
            else cc(amount -
    first_denomination(kinds_of_coins), kinds_of_coins)
                + cc(amount, kinds_of_coins - 1))
```

Adding base cases

```
def cc(amount, kinds_of_coins):
    return (1 if amount == 0
            else 0 if amount < 0 or kinds_of_coins == 0
            else cc(amount -
    first_denomination(kinds_of_coins), kinds_of_coins)
                + cc(amount, kinds_of_coins - 1))
```

Step 5

Test your program

The CS1101S 5-step Method of Problem Solving™

Step 1

Read the problem *very carefully*

The CS1101S 5-step Method of Problem Solving™

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Step 2

Play with examples

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Program using Python

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Program using Python

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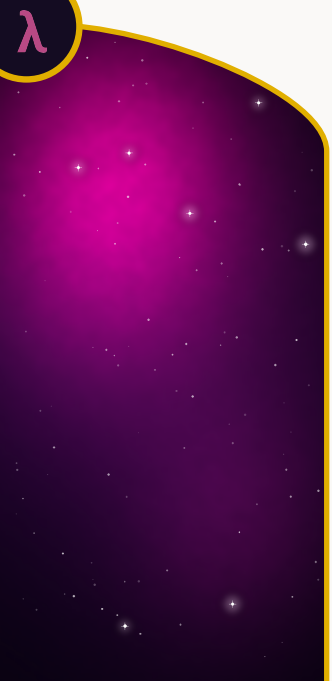

Passing functions to functions

```
def f(g, x):
    return g(x)

def g(y):
    return y + 1

print(f(g, 7))
```

Passing more functions to functions

```
def f(g, x):
    return g(g(x))

def g(y):
    return y + 1

print(f(g, 7))
```

We have seen this before!

Repeating the pattern n times

```
show(repeat_apply(make_cross, 4, rcross))
```

We have seen this before!

Repeating the pattern *n* times

```
show(repeat_apply(make_cross, 4, rcross))
```

Combining and transforming pictures

```
def transform_mosaic(p1, p2, p3, p4, transform):
    return transform(mosaic(p1, p2, p3, p4))
```

Look at this function (1.3.1)

```
def sum_integers(a, b):
    return 0 if a > b else a + sum_integers(a + 1, b)
```

...and this one

```
def cube(x):
    return x * x * x

def sum_skip_cubes(a, b):
    return 0 if a > b else cube(a) + sum_skip_cubes(a +
    2, b)
```

Abstraction (1.3.1)

```
def sum(a, b):
    return (0 if a > b
            else ⟨compute value with a⟩
                 + sum(⟨next value from a⟩, b))
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Abstraction (1.3.1)

```
def sum(a, b):
    return (0 if a > b
            else ⟨compute value with a⟩
                 + sum(⟨next value from a⟩, b))
```

in Python:

```
def sum(term, a, next, b):
    return 0 if a > b else term(a) + sum(term, next(a),
    next, b)
```

sum_integers **using** sum

```
def identity(x):
    return x

def plus_one(x):
    return x + 1

def sum_integers(a, b):
    return sum(identity, a, plus_one, b)
```

sum_skip_cubes **using** sum

```
def cube(x):
    return x * x * x

def plus_two(x):
    return x + 2

def sum_skip_cubes(a, b):
    return sum(cube, a, plus_two, b)
```

sum_skip_cubes **using** sum

```
def cube(x):
    return x * x * x

def plus_two(x):
    return x + 2

def sum_cubes(a, b):
    return sum(cube, a, plus_two, b)
```

Visibility

sum_skip_cubes **using** sum

```
def cube(x):
    return x * x * x

def plus_two(x):
    return x + 2

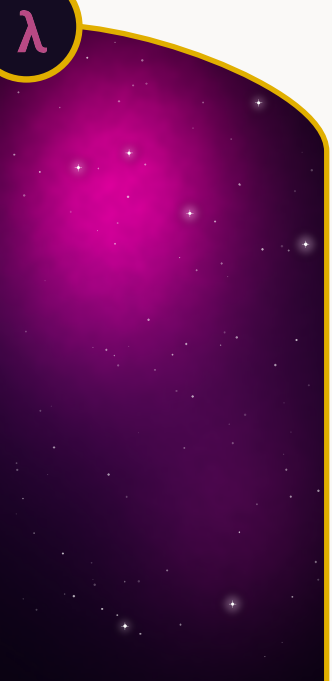
def sum_cubes(a, b):
    return sum(cube, a, plus_two, b)
```

Visibility

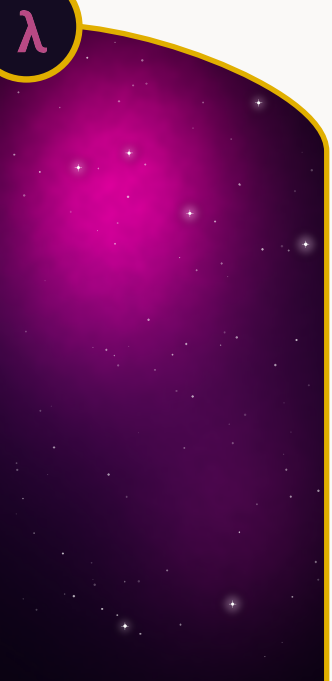
Can we "hide" `cube` and `plus_two` inside of `sum_skip_cubes`?

Yes, we can! (see also SICPy 1.1.8)

```
def sum_skip_cubes(a, b):
    def cube(x):
        return x * x * x
    def plus_two(x):
        return x + 2
    return sum(cube, a, plus_two, b)
```

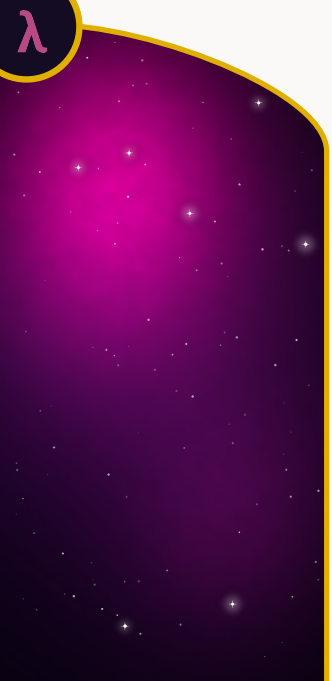


Now You See It, Now You Don't



Now You See It, Now You Don't

Let's tuck `fact_iter` in and see how many parameters it can shed along the way!



Now You See It, Now You Don't

Let's tuck `fact_iter` in and see how many parameters it can shed along the way!

Look for Check-In I3A in the Source Academy!

Another look at such local functions

```
def sum_skip_cubes(a, b):
    def cube(x):
        return x * x * x
    def plus_two(x):
        return x + 2
    return sum(cube, a, plus_two, b)
```

Another look at such local functions

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This is still quite verbose

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This is still quite verbose

Do we need all these words such as `def`, `return`?

Another look at such local functions

```
def sum_skip_cubes(a, b):
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    def plus_two(x):
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    return sum(cube, a, plus_two, b)
```

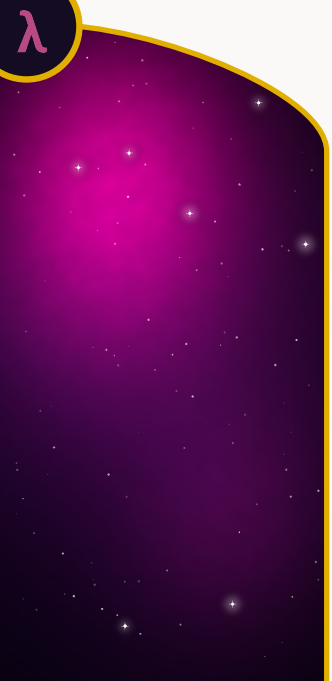
This is still quite verbose

Do we need all these words such as `def`, `return`?
Do we need to even give names to these functions?

No, we don't! Lambda expressions

```
def sum_skip_cubes(a, b):
    def cube(x):
        return x * x * x
    def plus_two(x):
        return x + 2
    return sum(cube, a, plus_two, b)
```

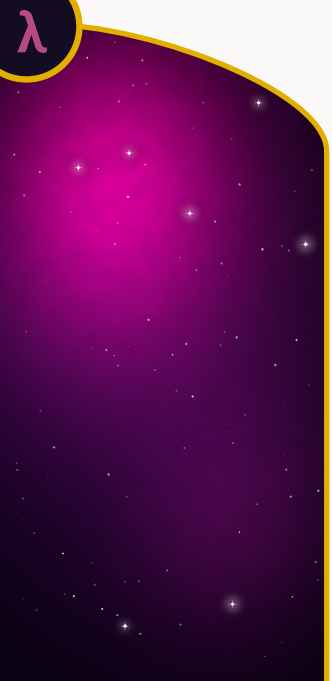
```
# instead just write:
def sum_skip_cubes(a, b):
    return sum(lambda x: x * x * x, a, lambda x: x + 2, b)
```



Lambda expressions (1.3.2)

New kind of expression

`lambda` *parameters* : *expression*



Lambda expressions (1.3.2)

New kind of expression

`lambda` *parameters* : *expression*

Meaning

The expression evaluates to a function value.
Function has given *parameters* and *expression* as its (only) return value; no `return` keyword needed.

Lambda expressions (1.3.2)

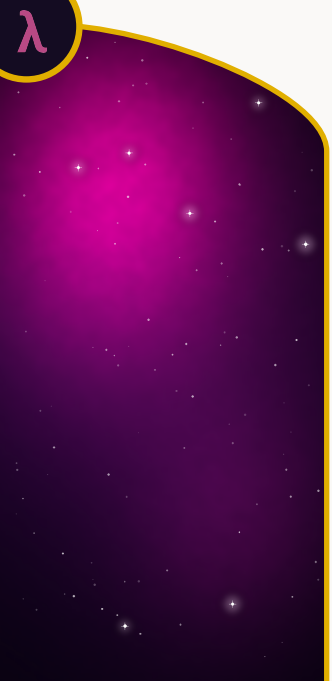
New kind of expression

`lambda` *parameters* `:` *expression*

Meaning

The expression evaluates to a function value.
Function has given *parameters* and *expression* as its (only) return value; no `return` keyword needed.

(yes, this is also why it's on our t-shirts)



We can have more than one parameter

Comma-separated, just like a function definition

`lambda` p_1`,` p_2`,` `...`, p_n `:` *expression*

We can have more than one parameter

Comma-separated, just like a function definition

`lambda` p_1`,` p_2`,` `...` `,` p_n `:` *expression*

```
add_three = lambda x, y, z: x + y + z

print(add_three(1, 2, 3))
```

An alternative syntax for function definition

```
def plus4(x):
    return x + 4
```

can be written as

```
plus4 = lambda x: x + 4
```

Returning Functions from Functions

```
def make_adder(x):
    def adder(y):
        return x + y
    return adder

adder_four = make_adder(4)
print(adder_four(6))
```

...or with the new lambda expressions

```
def make_adder(x):
    return lambda y: x + y

adder_four = make_adder(4)
print(adder_four(6))
```

Returning Functions from Functions

```
def make_adder(x):
    return lambda y: x + y

print((make_adder(4))(6))

# we can drop the parentheses:
#
# make_adder(4)(6)
```

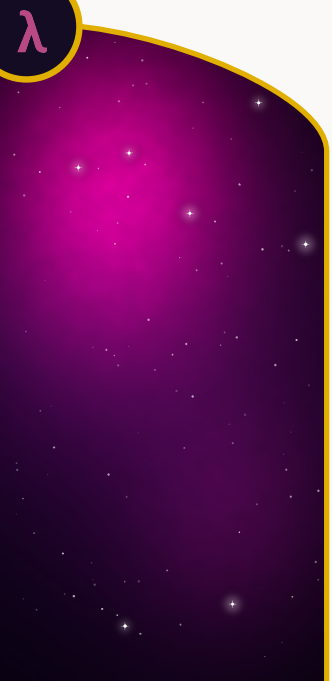
Returning Functions from Functions

```
def make_adder(x):
    return lambda y: x + y

adder_1 = make_adder(1)
adder_2 = make_adder(2)

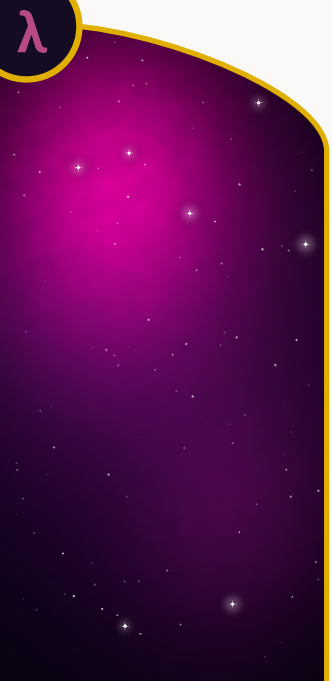
# adder_1(10) returns 11
print(adder_1(10))

# adder_2(20) returns 22
print(adder_2(20))
```



Summary of new constructs today

- Nested declaration assignment and function definition statements
- Conditional statements
- Lambda expressions



Scope of names: an example

```
z = 2
def f(g):
    z = 4
    return g(z)

print(f(lambda y: y + z))
```

Scope of names: an example

```
z = 2
def f(g):
    z = 4
    return g(z)

print(f(lambda y: y + z))
```

Questions about scope

What names are declared by this program?

Scope of names: an example

```
z = 2
def f(g):
    z = 4
    return g(z)

print(f(lambda y: y + z))
```

Questions about scope

What names are declared by this program?
Which declaration does each name occurrence refer to?

Scope of names: another example

```
x = 10
def square(x):
    return x * x

def addx(y):
    return y + x

print(square(x + 5) * addx(x + 20))
```

Questions about scope

Which declaration does each occurrence of `x` refer to?

Scope of names: yet another example

```
pi = 3.141592653589793

def circle_area_from_radius(r):
    pi = 22 / 7
    return pi * square(r)
```

Questions about scope

Which declaration does the occurrence of `pi` refer to?

Scope of names: `hypotenuse` example

```
def square(x):
    return x * x

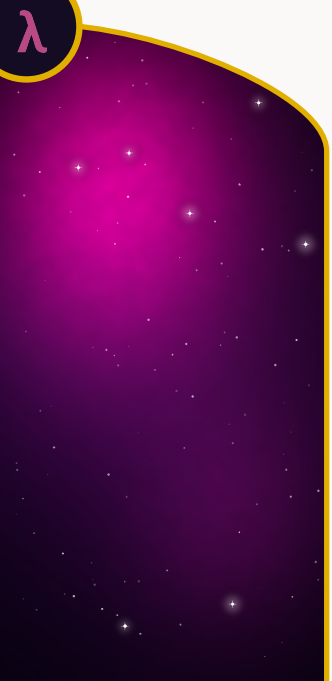
def hypotenuse(a, b):
    def sum_of_squares():
        return square(a) + square(b)
    return math_sqrt(sum_of_squares())
```

Scope of names: `hypotenuse` example

```
def square(x):
    return x * x

def hypotenuse(a, b):
    def sum_of_squares():
        return square(a) + square(b)
    return math_sqrt(sum_of_squares())
```

Names can refer to declarations outside of the immediately surrounding function definition.



Overview of scoping rules

Declarations mandatory

All names in Python must be declared.

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Forms of declaration

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2. Declaration assignments
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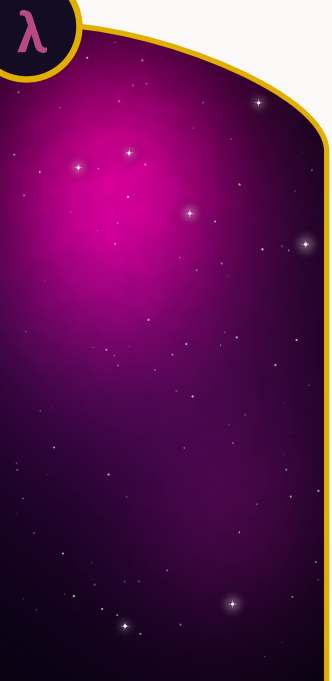
Scoping rule

A name occurrence refers to the *closest surrounding declaration* of the name.



Forms of Declaration

(1) Primitive names

The Python §1 pages tell us what names are primitive, e.g. `math_floor`.

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We can also `import` further primitive names from modules. For example, from the rune module:

```
from rune import heart, quarter_turn_right
```

Forms of Declaration

(2) Declaration assignments

The scope of a declaration assignment is the body of the *immediately surrounding function*, or the whole program, if there is none.

A simple example (see SICPy 1.3.2)

```
z = 3
def f():
    print(z)
    z = 4
f()
```

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Forms of Declaration

(2) Declaration assignments: what about branches?

```
z = 100
def f(x, y):
    if x > 0:
        z = x * y
    return z
print(f(3, 4))
```

Forms of Declaration

(2) Declaration assignments: what about branches?

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print(f(3, 4))
```

The declaration assignment `z = x * y` is inside the `if`, but the local `z` still shadows the global one throughout `f`.

Forms of Declaration

(2) Declaration assignments: scope vs. value

`z`'s *scope* covers all of `f`, but it only has a *value* once the assignment has actually run.

```
z = 100

def f(x, y):
    if x > 0:
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print(f(-3, 4))
```

Forms of Declaration

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```
z = 100

def f(x, y):
    if x > 0:
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    return z

print(f(-3, 4))
```

`f(-3, 4)` skips the `if`: `z` is still in scope but never gets a value (*not even the global `z` above*), so this is an error.

Forms of Declaration

(3) Parameters

The scope of the parameters of a lambda expression or function definition is the body of the function.

```
def f(x, y, z):
    ... x ... y ... z ...

lambda v, w, u: ... v ... w ... u ...
```

Forms of Declaration

(4) Function name

The scope of the function name of a function definition is as if the function was declared with a declaration assignment.

```
def f(x):
    ...
```

as if we wrote

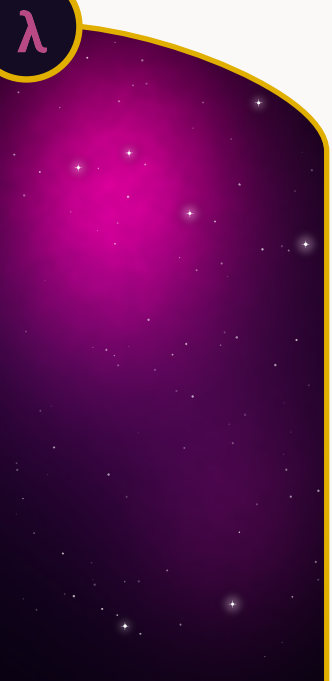
```
f = ...
```

Lexical scoping

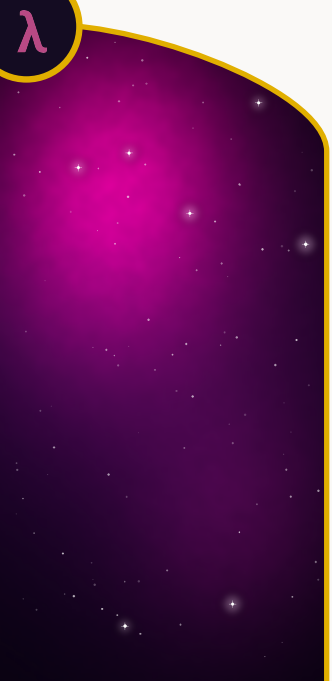
Scoping rule

A name occurrence refers to the *closest surrounding declaration* of the name.

```
def f(x):
    def g(w):
        x = ...
        y = x + 1
        return lambda x: ...x...
    return g(x)
```

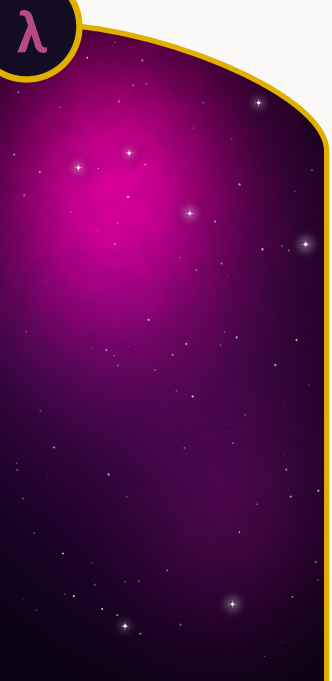


Important Ideas



Important Ideas

- *Hiding* can be a useful abstraction technique



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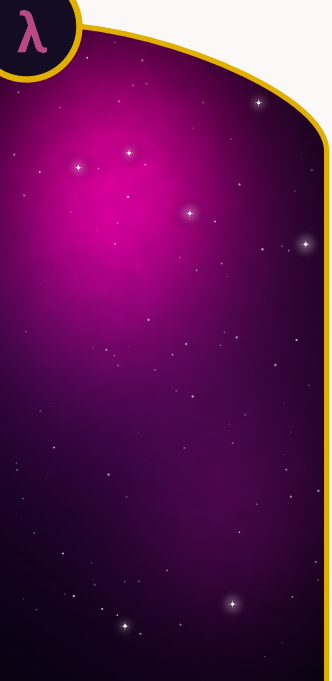

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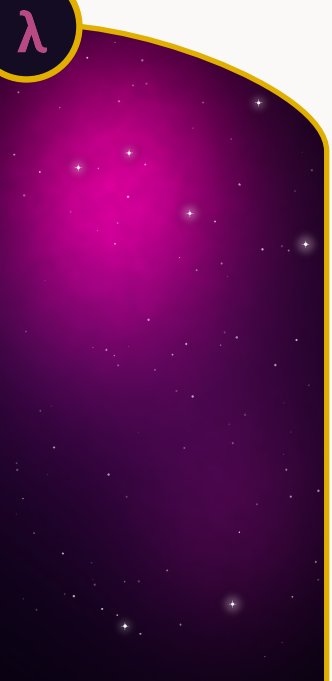

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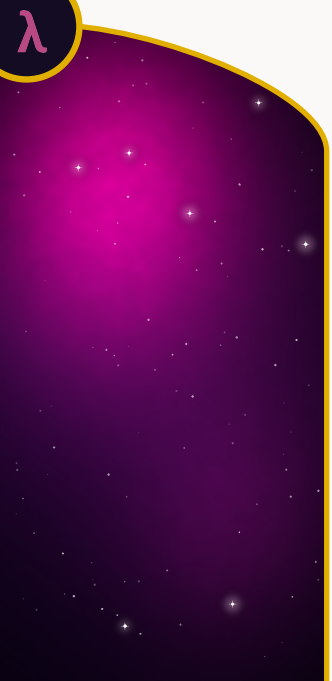

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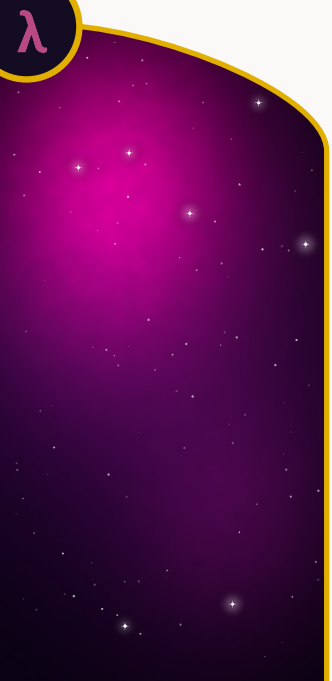

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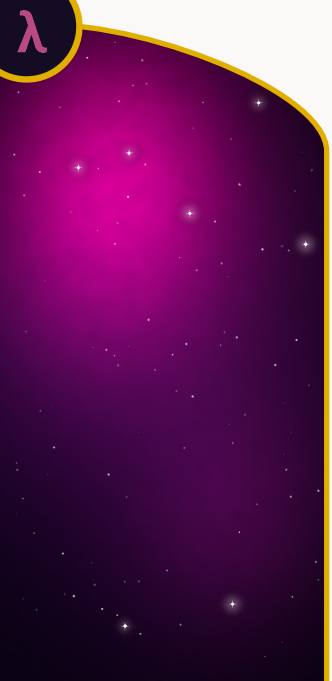
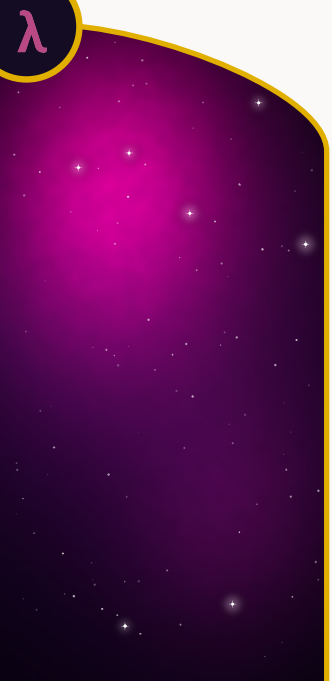


Concluding Unit 1



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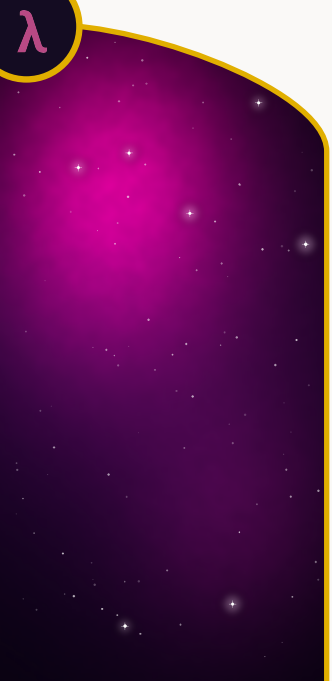

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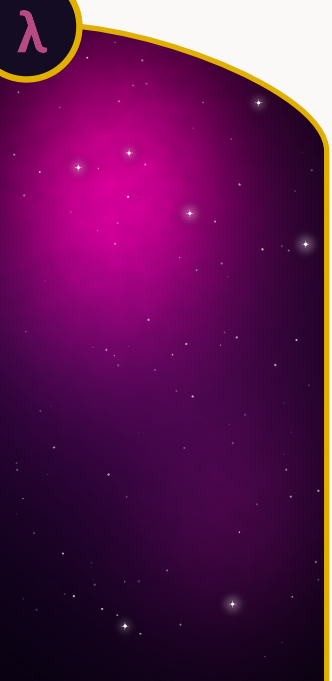

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